

Selected work samples

Metric design. Trustworthy data. Decisions that survive scrutiny.
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01 NEW CONTENT EXPERIENCE MEASUREMENT LAB

Does higher observed return justify expanding a new live experience?

The Python/SQL pipeline turns a product question into a traceable decision memo. It connects explicit eligibility rules, event quality, member-level metrics, a filterable dashboard, and an explanation of selection bias.

What is implemented. Seeded event generation; duplicate and conflicting-ID handling; event-time sessionization; first-day adoption and engagement; seven-day return; Wilson intervals; and descriptive standardization by pre-index activity.

The apparent advantage shrinks after accounting for audience mix

Raw return difference		+30.4 pp
Standardized difference		+4.6 pp

Observed in the simulation: 1,194 of 1,600 members have seven-day event-time maturity. Return is 64.0% among exposed members (445/695) and 33.7% among unexposed members (168/499). Another 406 members are excluded from all displayed member rates.

Interpretation. Frequent viewers are deliberately more likely to be exposed and to return. The simulator gives exposure no causal effect. Standardization removes much of the raw difference; the remaining gap is not evidence of incremental benefit.

Decision. Validate instrumentation, monitor playback-quality guardrails, then design a randomized exposure study. Observed return alone does not establish satisfaction or product value.

TECHNICAL EVIDENCE

Inspectable SQL, hand-calculated and failure-mode tests, reproducible CSV/SQLite outputs, and an offline HTML dashboard. Core execution uses the Python standard library; PDF export uses ReportLab.

Scope: synthetic data only; no Netflix or employer data. No completed A/B test, causal lift, production deployment, or distributed-scale claim. Source, tests and an implementation walkthrough accompany the project.

Reliable inputs. Explainable systems.

Two larger private applications complement the measurement lab. This summary reflects documentation review; their full test suites were not rerun. The public CSV and in-memory examples implement narrower workflows.

02 DATA RECONCILIATION WORKBENCH

Make migration exceptions visible before trusting the output.

Question. How can inconsistent business records be reconciled without silently accepting ambiguous matches or duplicating records on a rerun?

Demonstrated workflow. A local Python/Tkinter application validates a structured Excel import, normalizes customer, product and transaction data, and surfaces schema errors and possible duplicates. A human reviews ambiguous matches before accepted records are committed transactionally.

Evidence and relevance. Dry-run results, Excel/CSV/HTML/JSON exports and audit records make the transformation reviewable. Identical reruns are recognized. The analytics connection is trustworthy source data with explicit exceptions and reconciliation.

Boundary: fictional data and local execution. No paid-client deployment, verified business savings, security certification or production-scale claim.

03 RELIABLE EVENT-DRIVEN OPERATIONS BACKEND

Keep retries and recovery from corrupting system state.

Question. How can a system process repeated events and conflicting resource requests while preserving evidence of successful and failed operations?

Demonstrated workflow. A FastAPI/SQLAlchemy/SQLite service with a Streamlit dashboard models transactional resource allocation, idempotent events, artifact custody, and SHA-256-verified backup. Operational views expose failures and recovery states.

Evidence and relevance. Synthetic scenarios cover duplicate and out-of-order events, allocation conflicts, failed checksums and retry. Reliable event semantics and inspectable state are foundations for dependable analytical outputs.

Boundary: single-host synthetic system. Application-enforced audit controls are not tamper-proof or WORM storage. Multi-instance concurrency and production access controls require additional work.

REVIEW & PROVENANCE

The applications use fictional data. The measurement lab includes source, reproducible outputs, tests and a metric-review walkthrough. Private-application capabilities above are not claims about the narrower public examples. Source and walkthrough: github.com/bcrain1/portfolio/tree/main/content_measurement_lab.